

		$\approx 3010 \text{ m s}^{-1}$
11	B	Extension of spring with load of 6.0 N is 0.030 m. By extending a further 0.010 m, the additional elastic potential energy stored in the spring is the area under the F-x graph from $x = 0.030 \text{ m}$ to $x = 0.040 \text{ m}$. Additional elastic potential energy $= \frac{1}{2}(6 + 8)(0.010)$ $= 0.070 \text{ J}$
12	D	Option A is incorrect as Newton's 1st law states the resultant force acting